

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	V Deepa Dharshini	Reg. No.:	192525213
Section / Batch:	B.tech AIML	Date:	28-7-26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

CO2 - AT-1 Analytical Problem

Solving

Bubble Sort - Alternating High & Low Values

① Problem Interpretation:

Bubble Sort is a Brute Force sorting technique because it solves the sorting problem in the most straightforward way possible: It repeatedly scans the array & directly compares every pair adjacent elements, without any clever pre-processing or divide-and-conquer strategy. If two adjacent elements are out of order, it swaps them. This process is repeated pass after pass until no swaps are needed, "bubbling" the largest unsorted element to its correct position at the end of the array in each pass.

Data set : $[50, 10, 40, 20, 30, 5]$ ($n=6$)

(Worst-case input for bubble Sort)

Total Comparisons and Swaps

Pass	Comparisons	Swaps
1	5	5
2	4	3
3	3	1
4	2	1
5	1	1
Total	15	11

Total comparisons : $15 = \frac{n(n-1)}{2}$ [W.K.T, $n=6$, $\frac{6(5)}{2} = 15$]

Total swaps : 11 (out of maximum possible 15)

4) Time Complexity Analysis:

Best Case:

→ Array already sorted

→ Complexity : $O(n)$

Average Case:

→ Elements in random order

→ $O(n^2)$

Worst Case:

→ Array sorted in reverse order

→ $O(n^2)$

Space Complexity: $O(1)$

5) Verification of Final Output:

Final sorted array: $[5, 10, 20, 30, 40, 50]$

Checking: $5 < 10 < 20 < 30 < 40 < 50$

Correctly sorted in ascending order

6) Efficiency Interpretation on Alternating Values:

An alternating sequence like $[50, 10, 40, 20, 30, 5]$ is inefficient for bubble sort because many adjacent elements are out of order, resulting in frequent swaps (11 out of 15).

Since bubble sort moves elements only one position at a time, values far from their correct position require multiple passes, giving it an $O(n^2)$ time complexity on disordered data.

Code of Conduct

(192525213)

I, Deepa Dhareshini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deepa**RUBRICS FOR CALCULATION**

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____